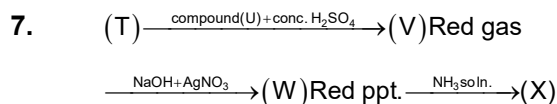
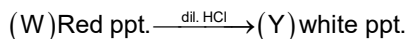


- When MnO_2 is fused with KOH in the presence of air, the product formed is :
 (A) purple colour KMnO_4 (B) green colour K_2MnO_4
 (C) colourless MnO_4^- (D) purple colour K_2MnO_4
- In which of the following species, the colour is due to charge transfer.
 (I) $[\text{Mn}(\text{OH})_4]^{2-}$ (II) MnO_4^{2-}
 (III) MnO_2 (IV) KMnO_4
 (A) I, II, III correct (B) II, IV correct (C) I, III correct (D) only IV correct
- XCl_2 (excess) + $\text{YCl}_2 \rightarrow \text{XCl}_4 + \text{Y}$;
 $\text{YO} \xrightarrow[\text{>400}^\circ]{\Delta} \frac{1}{2} \text{O}_2 + \text{Y}$, Ore of Y would be:
 (A) Siderite (B) Cinnabar (C) Malachite (D) Hornsilver
- Choose the correct code regarding Roasting process.
 (I) It is the process of heating ore in air to obtain the oxide
 (II) It is an exothermic process
 (III) It may be used for hydrated oxide and oxysalt ore to remove impurities
 (IV) It is used after the concentration of ore
 (A) I, II and III (B) I, II and IV
 (C) I, III and IV (D) I, II, III and IV
- Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.
 (i) In Goldschmidt thermite process aluminium acts as a reducing agent.
 (ii) Mg is extracted by electrolysis of aq. solution of MgCl_2
 (iii) Extraction of Pb is possible by carbon reduction method
 (iv) Red Bauxite is purified by Serpeck's process
 (A) T T T F (B) T F F T (C) F T T T (D) T F T F
- Which of the following statement is/are incorrect?
 (A) Hall and Heroult's method is used in extraction of Al .
 (B) Iron obtained from blast furnace is cast iron.
 (C) Wrought iron is obtained by puddling process.
 (D) None of these



T is an Orange crystalline compound which when burnt imparts violet color.



Identify (T) to (Z).

- (A) $T = \text{KMnO}_4$, $U = \text{HCl}$, $V = \text{Cl}_2$, $W = \text{HgI}_2$, $X = \text{Hg}(\text{NH}_2)\text{NO}_3$, $Y = \text{Hg}_2\text{Cl}_2$, $Z = \text{N}_2$
 (B) $T = \text{K}_2\text{Cr}_2\text{O}_7$, $U = \text{NH}_4\text{Cl}$, $V = \text{CrO}_2\text{Cl}_2$, $W = \text{Ag}_2\text{CrO}_4$, $X = [\text{Ag}(\text{NH}_3)_2]^+$, $Y = \text{AgCl}$, $Z = \text{NH}_3$
 (C) $T = \text{K}_2\text{CrO}_4$, $U = \text{KCl}$, $V = \text{CrO}_2\text{Cl}_2$, $W = \text{HgI}_2$, $X = \text{Na}_2\text{CrO}_4$, $Y = \text{BaCO}_3$, $Z = \text{NH}_4\text{Cl}$
 (D) $T = \text{K}_2\text{MnO}_4$, $U = \text{NaCl}$, $V = \text{CrO}_3$, $W = \text{AgNO}_2$, $X = (\text{NH}_4)_2\text{CrO}_4$, $Y = \text{CaCO}_3$, $Z = \text{SO}_2$
8. Metal-metal bonding is more frequent in 4d or 5d series than in 3d series due to:
- their greater enthalpies of atomisation
 - the large size of the orbitals which participates in the metal-metal bond formation
 - their ability to involve both ns and (n – 1) d electrons in the bond formation
 - the comparable size of 4d and 5d series elements

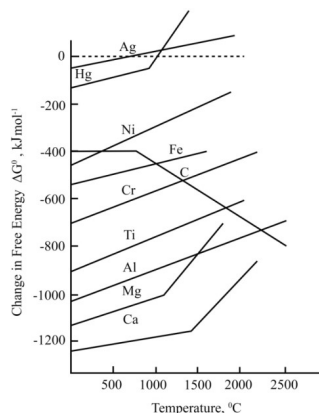
MCQ

9. Select correct statement regarding silver extraction process.
- When the lead-silver alloy is rich in silver, lead is removed by the cupellation process
 - When the lead-silver alloy is rich in lead, lead is removed by Parke's or pattinson's process
 - Zinc forms an alloy with lead, from which lead is separated by distillation
 - Zinc forms an alloy with silver, from which zinc is separated by distillation
10. Correct statements about froth floatation process used in metallurgy
- It is based on the difference in wettability of different minerals
 - Sodium ethyl xanthate is used as collector
 - NaCN is used as a depressant in the mixture of ZnS and PbS when ZnS forms soluble complex
 - This method is generally employed to dress the sulphide ores
11. Which of the following is true for calcination of a metal ore ?
- It makes the ore more porous.
 - The ore is heated to a temperature when fusion just begins.
 - Hydrated salts lose their water of crystallisation.
 - Impurities of S, As and Sb are removed in the form of their volatile oxides.
12. Which of the following statements regarding d-block elements are true?
- the colour of anhydrous CuSO_4 is blue
 - "splitting of silver" can be prevented by covering the surface of molten silver with charcoal
 - Iodine liberated in a reaction can be estimated by titration against a standard thiosulphate solution
 - Lanthanum is first element of third transition series.

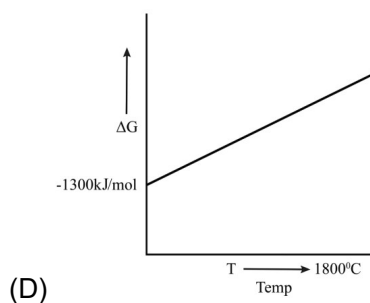
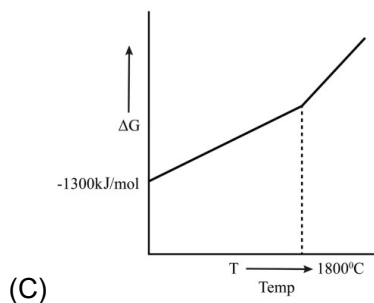
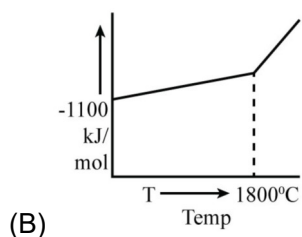
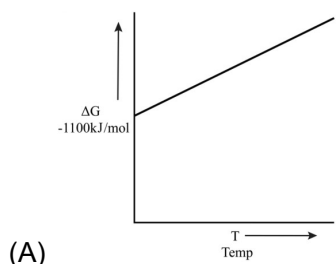
Comprehension

For a spontaneous reaction, the free energy change must be negative. $\Delta G = \Delta H - T\Delta S$, ΔH is the enthalpy change during the reaction. T is the absolute temperature, and ΔS is the change in entropy during the reaction. Consider a reaction such as the formation of an oxide $M + O_2 \rightarrow MO$. Dioxygen is used up in the course of this reaction. Gases have a more random structure (less ordered) than liquid or solids. Consequently, gases have a higher entropy than liquids and solids. In this reaction S (entropy or randomness) decreases, hence ΔS is negative. Thus, if the temperature is raised then $T\Delta S$ becomes more negative. Since, $T\Delta S$ is subtracted in the equation, then ΔG becomes less negative. Thus, the free energy change increases with the increase in temperature.

The free energy changes that occur when one mole of common reactant (in this case dioxygen) is used may be plotted graphically against temperature for a number of reactions of metals to their oxides. The following plot is called an Ellingham diagram for metal oxide. Understanding of Ellingham diagram is extremely important for the efficient extraction of metals.



13. For the conversion of $Ca(s)$ to $CaO(s)$ which of the following represent the ΔG vs T ?



14. Free energy change of Hg and Mg for the conversion to oxides the slope of ΔG vs. T has been changed above the boiling points of the given metal because
 (A) above the boiling point of the metal, entropy is increased
 (B) above the boiling point of the metal, entropy is decreased
 (C) above the boiling point of the metal, entropy change is equal to zero
 (D) All of these
15. Which of the following elements can be prepared by heating the oxide above 400°C ?
 (A) Hg (B) Mg (C) Fe (D) Al
16. As per the Ellingham diagram of oxides which of the following conclusion is true?
 (A) Al reduces Fe_2O_3 , whereas MgO cannot be reduced by Al at 1500°C
 (B) Fe reduces Al_2O_3 , whereas MgO cannot be reduced by Al at 1500°C
 (C) Al reduces Fe_2O_3 , whereas MgO cannot be reduced by Ca at 1500°C
 (D) Al can reduce both Fe_2O_3 and MgO to the corresponding metal at 1500°C

Comprehension

Magnesium is a valuable, light weight metal used as a structural material as well as in alloys, batteries, and in chemical synthesis. Although magnesium is plentiful in Earth's crust, it is mainly found in the sea water (after sodium). There is about 1.3 g of magnesium in every kilogram of sea water. The process for obtaining magnesium from sea water employs all three types of reactions, i.e., precipitation, acid-base, and redox reactions.

17. Precipitation reaction involves formation of
 (A) insoluble MgCO_3 by adding Na_2CO_3
 (B) insoluble $\text{Mg}(\text{OH})_2$ by adding $\text{Ca}(\text{OH})_2$
 (C) insoluble in MgSO_4 by adding Na_2SO_4
 (D) insoluble MgCl_2 by adding NaCl
18. Acid-base reaction involves reaction between
 (A) MgCO_3 and HCl (B) $\text{Mg}(\text{OH})_2$ and H_2SO_4
 (C) $\text{Mg}(\text{OH})_2$ and HCl (D) MgCO_3 and H_2SO_4
19. Redox reaction takes place
 (in the extraction of Mg)
 (A) in the electrolytic cell when fused MgCl_2 is subjected to electrolysis
 (B) when fused MgCO_3 is heated
 (C) when fused MgCO_3 is strongly heated
 (D) none of the above

INTEGER

20. How many of the following metals can be extracted by carbon reduction method?
 Na, Al, Mg, Sn, Fe, Pb, Zn, Ca
21. No. of following reagents in which ppt A is soluble is :
 $\text{AgNO}_3 + \text{NaCl} \longrightarrow \text{A}$
 (i) aq NH_3 (ii) aq. KCN (iii) $\text{Na}_2\text{S}_2\text{O}_3$
 (iv) Conc. HCl (v) aq. NaOH (iv) aq. KI

22. Column-I and Column-II contains four entries each. Entries of column-I are to be match with some entries of column-II. Each entry of column-I may have the matching with one or more than one entries of column-II.

Column-I

- (A) Hydrometallurgy applied in commercial extraction of metal
 (B) carbon reduction applied in commercial extraction in metal
 (C) Aqueous salt solution is used in electrolytic Refining method
 (D) Metal present in anode mud of refining of crude copper

Column-II

- (p) Ag
 (q) Zn
 (r) Sn
 (s) Au
 (t) Cu

- (A) $A \rightarrow P, S; B \rightarrow Q; C \rightarrow P, Q; D \rightarrow P, S$
 (B) $A \rightarrow P, S; B \rightarrow Q, R; C \rightarrow P, S; D \rightarrow P, Q, R, S, T$
 (C) $A \rightarrow Q, R; B \rightarrow P, S; C \rightarrow P, Q, R, S, T; D \rightarrow P, S$
 (D) $A \rightarrow P, S, T; B \rightarrow Q, R; C \rightarrow P, Q, R, S, T; D \rightarrow P, S$

23. Column-I contains four statements following reason and Column-II consists of four options P, Q, R, S Answer the following :

- P $\rightarrow$ If both statement and reason are true and reason is correct explanation of statement.
 Q $\rightarrow$ If both statement and reason are true and reason is not correct explanation of statement.
 R $\rightarrow$ If statement is correct and reason is incorrect.
 S $\rightarrow$ If both statement and reason are incorrect.

Column-I

- (A) **Statement** : The reaction of oxalic acid with acidified KMnO_4 is first slow and then speeds up by itself.
Reason : KMnO_4 decomposes into MnO_2 in sunlight.
 (B) **Statement** : Anhydrous ZnCl_2 can't be made by heating $\text{ZnCl}_2 \cdot 2\text{H}_2\text{O}$.
Reason : It undergoes hydrolysis to produce Zn(OH)_2 and HCl .
 (C) **Statement** : KMnO_4 is not used as a primary standard substance
Reason : It is deliquescent in nature.
 (D) **Statement** : $\text{K}_2\text{Cr}_2\text{O}_7$ has orange colour due to polarisation.
Reason : In dichromate ion all Cr—O bonds are identical.

Column-II

- (P)
 (Q)
 (R)
 (S)

24. A white substance (A) reacts with dilute H_2SO_4 to produce a colourless suffocating gas (B) and a colourless solution (C). The reaction of gas (B) with potassium iodate and starch solution produces a blue colour solution. Aqueous solution of (A) gives a white precipitate with BaCl_2 solution which is soluble in dilute HCl . Addition of aqueous NH_3 or NaOH to (C) produces first a precipitate which dissolves in excess of the respective reagent to produce a clear solution. Similarly, addition of excess of potassium ferrocyanide to (C) produces a precipitate (D) which also dissolves in aqueous NaOH giving a clear solution. Identify (A), (B), (C) and (D). Write the equations of the reactions involved.
25. Explain the large difference in melting point of Cr(1920°C) and Zn(420°C).